

Database

Topic 1: Database Fundamentals

Q1. What is a database?

Answer: A database is an organized collection of data that can be stored, managed, and retrieved efficiently.

Simple Explanation: It stores related information in a structured way.

Example: A college database can store student names, IDs, courses, and marks.

Q2. What is a DBMS?

Answer: DBMS stands for Database Management System. It is software used to create, manage, and access databases.

Simple Explanation: A DBMS helps users store, update, and retrieve data.

Example: MySQL is a popular DBMS.

Q3. What is the difference between data and information?

Answer: Data is raw facts, while information is processed data that has meaning.

Simple Explanation: Data becomes useful when it is organized and interpreted.

Example: 85 is data; Student scored 85 marks is information.

Q4. What are the advantages of using a database?

Answer: Databases provide organized storage, easy data retrieval, security, consistency, and reduced duplication.

Simple Explanation: They make managing large amounts of data easier.

Example: A bank can manage millions of customer records efficiently.

Q5. What is a relational database?

Answer: A relational database stores data in tables containing rows and columns.

Simple Explanation: Related tables can be connected using keys.

Example: MySQL and PostgreSQL are relational database systems.

Q6. What is a table in a database?

Answer: A table is a structured collection of related data arranged in rows

and columns.

Simple Explanation: Tables are used to store specific types of information.

Example: A Students table can store student details.

Q7. What is a row in a database table?

Answer: A row represents a single record in a table.

Simple Explanation: Each row contains information about one entity.

Example: One row in Students may represent one student.

Q8. What is a column in a database table?

Answer: A column represents a specific attribute of the data.

Simple Explanation: It defines what type of information is stored.

Example: Name, Age, and Email can be columns.

Q9. What is a record?

Answer: A record is a complete set of related data stored in a row.

Simple Explanation: A record usually describes one entity.

Example: A student's ID, name, and email together form a record.

Q10. What is a database schema?

Answer: A database schema is the logical structure or design of a database.

Simple Explanation: It defines tables, columns, relationships, and constraints.

Example: A schema may define Students, Courses, and Marks tables.

Q11. What is SQL?

Answer: SQL stands for Structured Query Language and is used to interact with relational databases.

Simple Explanation: SQL allows users to create, retrieve, update, and delete data.

Example: `SELECT * FROM Students;`

Q12. What is MySQL?

Answer: MySQL is a popular open-source relational database management system that uses SQL.

Simple Explanation: It is used to store and manage structured data.

Example: A website can use MySQL to store user accounts.

Q13. What is PostgreSQL?

Answer: PostgreSQL is an open-source relational database management system known for advanced features and reliability.

Simple Explanation: It supports complex queries and data types.

Example: A large application can use PostgreSQL for its backend database.

Q14. What is a database server?

Answer: A database server is a system that stores and manages databases and processes database requests.

Simple Explanation: It receives requests from applications and returns data.

Example: A web application may send SQL queries to a database server.

Q15. What is data redundancy?

Answer: Data redundancy means storing the same data unnecessarily in multiple places.

Simple Explanation: Repeated data can waste storage and cause inconsistencies.

Example: Storing the same customer address in several tables unnecessarily.

Q16. What is data consistency?

Answer: Data consistency means data remains accurate and uniform across the database.

Simple Explanation: The same information should not have conflicting values.

Example: A customer's phone number should be consistent wherever it is stored.

Q17. What is data integrity?

Answer: Data integrity ensures that data remains accurate, valid, and reliable.

Simple Explanation: Database rules prevent invalid data from being stored.

Example: A student ID should not be duplicated when it must be unique.

Q18. What is a database administrator?

Answer: A database administrator (DBA) manages, secures, monitors, and maintains databases.

Simple Explanation: The DBA ensures the database works correctly and securely.

Example: A DBA may create user permissions and perform backups.

Q19. What is a database model?

Answer: A database model defines how data is organized, stored, and related.

Simple Explanation: It provides a structure for designing databases.

Example: Relational, hierarchical, and network models are database models.

Q20. What is an entity in a database?

Answer: An entity is a real-world object or concept about which data is stored.

Simple Explanation: Entities become important objects in database design.

Example: Student, Employee, and Product can be entities.

Topic 2: SQL Basics & Queries

Q21. What is the purpose of the SELECT statement?

Answer: The SELECT statement retrieves data from one or more tables.

Simple Explanation: It is used to read information from a database.

Example: SELECT name FROM Students;

Q22. What is the purpose of the INSERT statement?

Answer: INSERT is used to add new records to a table.

Simple Explanation: It adds new data to the database.

Example: INSERT INTO Students VALUES (1, 'John');

Q23. What is the purpose of the UPDATE statement?

Answer: UPDATE is used to modify existing records.

Simple Explanation: It changes data that is already stored.

Example: UPDATE Students SET name='David' WHERE id=1;

Q24. What is the purpose of the DELETE statement?

Answer: DELETE removes existing records from a table.

Simple Explanation: It permanently removes selected rows.

Example: DELETE FROM Students WHERE id=5;

Q25. What is the CREATE TABLE statement?

Answer: CREATE TABLE is used to create a new database table.

Simple Explanation: It defines columns and their data types.

Example: CREATE TABLE Students (id INT, name VARCHAR(50));

Q26. What is the ALTER TABLE statement?

Answer: ALTER TABLE is used to modify the structure of an existing table.

Simple Explanation: It can add, modify, or remove columns.

Example: ALTER TABLE Students ADD email VARCHAR(100);

Q27. What is the DROP TABLE statement?

Answer: DROP TABLE permanently removes a table and its data.

Simple Explanation: Both the table structure and stored records are deleted.

Example: DROP TABLE Students;

Q28. What is the WHERE clause?

Answer: WHERE filters records based on a specified condition.

Simple Explanation: It selects only records that match the condition.

Example: SELECT * FROM Students WHERE age > 18;

Q29. What is the ORDER BY clause?

Answer: ORDER BY sorts query results in ascending or descending order.

Simple Explanation: It organizes the output based on a column.

Example: SELECT * FROM Students ORDER BY name ASC;

Q30. What is the DISTINCT keyword?

Answer: DISTINCT removes duplicate values from query results.

Simple Explanation: It displays only unique values.

Example: SELECT DISTINCT city FROM Students;

Q31. What is the LIKE operator?

Answer: LIKE is used to search for a specified pattern in text values.

Simple Explanation: It helps find partially matching strings.

Example: SELECT * FROM Students WHERE name LIKE 'A%';

Q32. What does the % wildcard represent in SQL?

Answer: % represents zero or more characters in a pattern.

Simple Explanation: It is commonly used with LIKE.

Example: LIKE 'A%' finds names beginning with A.

Q33. What does the _ wildcard represent in SQL?

Answer: _ represents exactly one character in a pattern.

Simple Explanation: It helps match a specific character position.

Example: LIKE 'J_hn' can match John.

Q34. What is the IN operator?

Answer: IN checks whether a value matches any value in a specified list.

Simple Explanation: It avoids writing multiple OR conditions.

Example: WHERE city IN ('Chennai','Madurai');

Q35. What is the BETWEEN operator?

Answer: BETWEEN selects values within a specified range.

Simple Explanation: It is useful for numbers, dates, and other ordered values.

Example: WHERE age BETWEEN 18 AND 25;

Q36. What is the AND operator in SQL?

Answer: AND returns records only when all specified conditions are true.

Simple Explanation: It combines multiple conditions.

Example: WHERE age > 18 AND city = 'Chennai';

Q37. What is the OR operator in SQL?

Answer: OR returns records when at least one specified condition is true.

Simple Explanation: It allows alternative conditions.

Example: WHERE city='Chennai' OR city='Madurai';

Q38. What is the NOT operator?

Answer: NOT reverses a condition in an SQL query.

Simple Explanation: It selects records that do not match the condition.

Example: WHERE NOT city='Chennai';

Q39. What is NULL in SQL?

Answer: NULL represents a missing, unknown, or unavailable value.

Simple Explanation: NULL is different from zero or an empty string.

Example: A student may have a NULL value for an optional phone number.

Q40. How do you check for NULL values?

Answer: Use IS NULL or IS NOT NULL.

Simple Explanation: Normal comparison operators should not be used to check NULL.

Example: SELECT * FROM Students WHERE email IS NULL;

Q41. What is an alias in SQL?

Answer: An alias is a temporary name given to a table or column in a query.

Simple Explanation: It makes query output or queries easier to read.

Example: SELECT name AS StudentName FROM Students;

Q42. What is the LIMIT clause?

Answer: LIMIT restricts the number of rows returned by a query in systems that support it.

Simple Explanation: It is useful when only a small number of results are needed.

Example: SELECT * FROM Students LIMIT 5;

Q43. What is the difference between DELETE and DROP?

Answer: DELETE removes records, while DROP removes the entire database object.

Simple Explanation: DELETE can remove selected rows; DROP removes

the table itself.

Example: DELETE FROM Students; removes rows, while DROP TABLE Students; removes the table.

Q44. What is the difference between DELETE and TRUNCATE?

Answer: DELETE removes rows and can use conditions, while TRUNCATE removes all rows quickly.

Simple Explanation: TRUNCATE is generally used when all table data must be cleared.

Example: TRUNCATE TABLE Students;

Q45. What is the purpose of comments in SQL?

Answer: SQL comments are used to add explanations or notes to SQL code.

Simple Explanation: Comments help developers understand queries.

Example: -- Retrieve all students

Topic 3: Keys, Constraints & Relationships

Q46. What is a primary key?

Answer: A primary key uniquely identifies each record in a table.

Simple Explanation: It cannot contain duplicate values.

Example: student_id can be a primary key.

Q47. What is a foreign key?

Answer: A foreign key is a column that references a key in another table.

Simple Explanation: It creates a relationship between tables.

Example: course_id in Students can reference Courses.

Q48. What is a candidate key?

Answer: A candidate key is a column or combination of columns that can uniquely identify records.

Simple Explanation: One candidate key is selected as the primary key.

Example: Student ID and email may both uniquely identify a student.

Q49. What is a composite key?

Answer: A composite key is a key made from two or more columns.

Simple Explanation: Multiple columns together uniquely identify a record.

Example: student_id + course_id can form a composite key.

Q50. What is a unique key?

Answer: A UNIQUE constraint ensures that values in a column are not duplicated.

Simple Explanation: It prevents duplicate values.

Example: Email addresses can be made UNIQUE.

Q51. What is the NOT NULL constraint?

Answer: NOT NULL prevents a column from storing NULL values.

Simple Explanation: It makes the field mandatory.

Example: A student name can be defined as NOT NULL.

Q52. What is the DEFAULT constraint?

Answer: DEFAULT provides an automatic value when no value is supplied.

Simple Explanation: It avoids leaving certain fields empty.

Example: status VARCHAR(20) DEFAULT 'Active'.

Q53. What is the CHECK constraint?

Answer: CHECK ensures that column values satisfy a specified condition.

Simple Explanation: It prevents invalid values.

Example: CHECK (age >= 18).

Q54. What is referential integrity?

Answer: Referential integrity ensures that foreign key values correctly reference existing records.

Simple Explanation: It prevents invalid relationships between tables.

Example: A course ID in Enrollments must exist in Courses.

Q55. What is a relationship in a database?

Answer: A relationship describes how records in different tables are connected.

Simple Explanation: Relationships help organize related data.

Example: Students can be related to Courses.

Q56. What is a one-to-one relationship?

Answer: A one-to-one relationship means one record in one table is related to one record in another table.

Simple Explanation: Each entity has at most one related entity.

Example: One person may have one passport record.

Q57. What is a one-to-many relationship?

Answer: A one-to-many relationship means one record can be related to many records.

Simple Explanation: It is one of the most common database relationships.

Example: One department can have many employees.

Q58. What is a many-to-many relationship?

Answer: A many-to-many relationship means many records in one table can relate to many records in another table.

Simple Explanation: It is usually implemented using a junction table.

Example: Students can enroll in many courses.

Q59. What is a junction table?

Answer: A junction table connects two tables involved in a many-to-many relationship.

Simple Explanation: It stores foreign keys from both related tables.

Example: Enrollment(student_id, course_id).

Q60. Can a table have more than one candidate key?

Answer: Yes, a table can have multiple candidate keys.

Simple Explanation: Any candidate key can potentially be selected as the primary key.

Example: Student ID and email may both be candidate keys.

Q61. Can a table have multiple foreign keys?

Answer: Yes, a table can contain multiple foreign keys.

Simple Explanation: Each foreign key can reference a different table.

Example: An Orders table may reference Customers and Products.

Q62. Can a primary key contain NULL values?

Answer: No, a primary key cannot contain NULL values.

Simple Explanation: Every record must have a valid unique identifier.

Example: student_id must always have a value.

Q63. What is entity integrity?

Answer: Entity integrity ensures that every record has a valid and unique primary key.

Simple Explanation: It prevents unidentified or duplicate records.

Example: Two students cannot have the same primary key.

Q64. What happens when a foreign key references a non-existing record?

Answer: The database generally rejects the operation when referential integrity is enforced.

Simple Explanation: This prevents invalid relationships.

Example: An invalid customer_id cannot be inserted into an Orders table.

Q65. Why are constraints important in databases?

Answer: Constraints enforce rules that keep data accurate, valid, and consistent.

Simple Explanation: They prevent common data-entry errors.

Example: NOT NULL can prevent a required student name from being empty.

Topic 4: Joins, Functions & Advanced SQL

Q66. What is a JOIN in SQL?

Answer: A JOIN combines related data from two or more tables.

Simple Explanation: It retrieves information stored across multiple tables.

Example: Joining Students with Courses using course_id.

Q67. What is an INNER JOIN?

Answer: INNER JOIN returns only records that have matching values in both tables.

Simple Explanation: Non-matching records are excluded.

Example: Students with valid course records can be retrieved using INNER JOIN.

Q68. What is a LEFT JOIN?

Answer: LEFT JOIN returns all records from the left table and matching records from the right table.

Simple Explanation: Unmatched right-side values become NULL.

Example: Display all students even if they have no course.

Q69. What is a RIGHT JOIN?

Answer: RIGHT JOIN returns all records from the right table and matching records from the left table.

Simple Explanation: Unmatched left-side values become NULL.

Example: Display all courses even if no student has enrolled.

Q70. What is a FULL OUTER JOIN?

Answer: FULL OUTER JOIN returns matching and non-matching records from both tables.

Simple Explanation: It combines the results of left and right joins.

Example: Show all students and all courses, including unmatched records.

Q71. What is a self join?

Answer: A self join joins a table with itself.

Simple Explanation: It is useful when records in the same table are related.

Example: An Employee table can be joined to itself to show managers and employees.

Q72. What is a CROSS JOIN?

Answer: CROSS JOIN produces every possible combination of rows from two tables.

Simple Explanation: Each row from one table is paired with every row from the other.

Example: Combining every product with every available color.

Q73. What is the COUNT() function?

Answer: COUNT() returns the number of rows or non-NULL values.

Simple Explanation: It is used to count records.

Example: SELECT COUNT(*) FROM Students;

Q74. What is the SUM() function?

Answer: SUM() calculates the total of numeric values.

Simple Explanation: It adds values in a selected column.

Example: SELECT SUM(salary) FROM Employees;

Q75. What is the AVG() function?

Answer: AVG() calculates the average of numeric values.

Simple Explanation: It returns the mean value.

Example: SELECT AVG(mark) FROM Students;

Q76. What is the MAX() function?

Answer: MAX() returns the highest value in a column.

Simple Explanation: It finds the maximum value.

Example: SELECT MAX(mark) FROM Students;

Q77. What is the MIN() function?

Answer: MIN() returns the lowest value in a column.

Simple Explanation: It finds the minimum value.

Example: SELECT MIN(mark) FROM Students;

Q78. What is GROUP BY?

Answer: GROUP BY groups rows with the same values for aggregate calculations.

Simple Explanation: It is commonly used with COUNT, SUM, and AVG.

Example: Count students in each department.

Q79. What is the HAVING clause?

Answer: HAVING filters grouped results after GROUP BY.

Simple Explanation: It is useful for conditions on aggregate results.

Example: Show departments having more than 10 students.

Q80. What is the difference between WHERE and HAVING?

Answer: WHERE filters rows before grouping, while HAVING filters groups after grouping.

Simple Explanation: WHERE works on individual rows; HAVING works on grouped results.

Example: WHERE filters students by age; HAVING filters departments by student count.

Q81. What is a subquery?

Answer: A subquery is a query written inside another SQL query.

Simple Explanation: It allows one query to use the result of another query.

Example: Find students whose marks are above the average.

Q82. What is a correlated subquery?

Answer: A correlated subquery depends on values from the outer query.

Simple Explanation: It is evaluated in relation to each outer row.

Example: Find employees earning more than the average salary in their department.

Q83. What is a view?

Answer: A view is a virtual table based on the result of an SQL query.

Simple Explanation: It can simplify complex queries and restrict access to data.

Example: A view can display only selected employee information.

Q84. What is an index?

Answer: An index is a database structure that improves the speed of data retrieval.

Simple Explanation: It helps the database find records faster.

Example: An index on email can speed up user searches.

Q85. What is a stored procedure?

Answer: A stored procedure is a precompiled group of SQL statements stored in the database.

Simple Explanation: It can be executed repeatedly for common operations.

Example: A procedure can generate a monthly sales report.

Topic 5: Normalization, Transactions & Security

Q86. What is normalization?

Answer: Normalization is the process of organizing data to reduce

redundancy and improve data integrity.

Simple Explanation: It divides data into related tables logically.

Example: Separating customer and order information into different tables.

Q87. What is First Normal Form (1NF)?

Answer: 1NF requires each column to contain atomic values and each record to be unique.

Simple Explanation: A cell should not contain multiple separate values.

Example: Store individual phone numbers separately instead of 9876, 8765 in one cell.

Q88. What is Second Normal Form (2NF)?

Answer: 2NF requires 1NF and removes partial dependencies on a composite key.

Simple Explanation: Non-key attributes must depend on the entire key.

Example: In a student-course table, course details should not depend only on student ID.

Q89. What is Third Normal Form (3NF)?

Answer: 3NF requires 2NF and removes transitive dependencies.

Simple Explanation: Non-key attributes should depend only on the primary key.

Example: Store department details in a separate Department table.

Q90. What is denormalization?

Answer: Denormalization intentionally adds redundancy to improve query performance.

Simple Explanation: It can reduce the number of joins required.

Example: Storing a frequently used customer name directly in an order summary.

Q91. What is a transaction?

Answer: A transaction is a group of database operations treated as one logical unit of work.

Simple Explanation: Either the required operations succeed together or

changes can be rolled back.

Example: Transferring money between two bank accounts.

Q92. What does ACID stand for?

Answer: ACID stands for Atomicity, Consistency, Isolation, and Durability.

Simple Explanation: These properties help ensure reliable database transactions.

Example: A bank transfer should complete safely without losing money.

Q93. What is Atomicity?

Answer: Atomicity means a transaction is completed entirely or not completed at all.

Simple Explanation: Partial transaction changes should not remain after failure.

Example: Both debit and credit operations must succeed in a transfer.

Q94. What is Consistency in ACID?

Answer: Consistency ensures that a transaction takes the database from one valid state to another.

Simple Explanation: Database rules remain satisfied after a transaction.

Example: A transaction cannot create an invalid foreign key.

Q95. What is Isolation in ACID?

Answer: Isolation ensures that concurrent transactions do not improperly interfere with each other.

Simple Explanation: Each transaction behaves as if it is safely separated from others.

Example: Two users updating the same account should not produce incorrect results.

Q96. What is Durability in ACID?

Answer: Durability ensures that committed transaction changes remain saved even after a system failure.

Simple Explanation: Once confirmed, data should not be lost unexpectedly.

Example: A completed bank transaction remains recorded after a server restart.

Q97. What is COMMIT in SQL?

Answer: COMMIT permanently saves the changes made by a transaction.

Simple Explanation: It confirms the transaction.

Example: COMMIT;

Q98. What is ROLLBACK in SQL?

Answer: ROLLBACK reverses changes made during the current transaction that have not been committed.

Simple Explanation: It is used when a transaction needs to be cancelled.

Example: ROLLBACK;

Q99. What is database security?

Answer: Database security protects data from unauthorized access, modification, or destruction.

Simple Explanation: It uses authentication, authorization, encryption, and other controls.

Example: Only authorized employees can access customer records.

Q100. What is database backup?

Answer: A database backup is a copy of database data stored for recovery purposes.

Simple Explanation: Backups help restore data after failures or accidental deletion.

Example: A company can restore its database from a recent backup after a server failure.